

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO1: Apply lexical analysis for token specification and recognition using LEX tool. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

**Duration :60 Mins
On Class
Total Marks:50**

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I Arunprasath R(192524077)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Arunprasath R

Annexure A

ANALYTICAL PROBLEM SOLVING

CO1 - Analytical Assessments

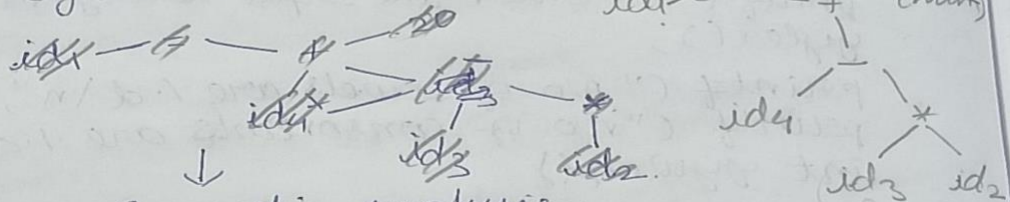
1) $w = x * y - z + 20$

↓
lexical Analysis.

$\langle id_1, w \rangle, \langle = \rangle, \langle id_2, x \rangle, \langle * \rangle, \langle id_3, y \rangle, \langle - \rangle,$
 $\langle id_4, z \rangle, \langle + \rangle, \langle 20 \rangle$

$id_1 = id_2 * id_3 - id_4 + 20$

↓
Syntax Analysis.



↓
Semantic Analysis.

The semantic analysis of this is same as syntax analysis.

↓
Intermediate code generator

$t_1 = id_2 * id_3$

$t_2 = t_1 - id_4$

$t_3 = t_2 + 20.0$

$id_1 = t_3$

↓
code optimization generator

$t_1 = id_2 * id_3$

$id_1 = (t_1 - id_4) + 20.0$

↓
code generator.

mov id_2, R_1

mov id_3, R_2

MUL R_2, R_1

Symbol table

x	...
y	...
z	...
w	...
.	...

SUB ~~R1~~ id1, R1
 ADD ~~R1~~, #20.0, R1
 MOV R1, id1.

2) Construct appropriate R.E for subset of $\{0,1\}^*$.

- (i) All strings beginning & ends with 1.
- (ii) All string contains sub-string 01.
- (iii) All string with exactly one 0.
- (iv) " " " at least two 1's.
- (v) " " " that do not end with 0.

Ans:

(i) $1(0+1)^*1$

(ii) $(0+1)^*01(0+1)^*$

(iii) 1^*01^*

(iv) $(0+1)^*1(0+1)^*1(0+1)^*$

(v) $(0+1)^*1 + \epsilon$

(3)

(i) $(\epsilon + aa^*)^* \Rightarrow [a^+]^* \quad [\because \epsilon + r^* = (r^*)^* = r^*]$
 $= a^*$ $\therefore (r^*)^* = r^*$

$\Rightarrow$ strings = $\{\epsilon, a, aa, aaa, aaaa, \dots\}$

$\Rightarrow$ Any non-negative integer count of 'a's.'

(ii) $(a^*b^*)^+ abba$

$\Rightarrow$ strings: ~~$\{ab, aab, aabb, aabab, aababb, \dots\}$~~

$\Rightarrow (a^*b^*)^+ = (a+b)^+$
 $= (a+b)^+ abba$

$L = abba$

(iii) $(a^*b^*)^+ abba$

$(a^* + b^*)^+ abba = (a+b)^+ abba$
 $L = abba$

(iv) $(a+ab)^*$

$L = \{\epsilon, a, aa, ab, aab, aba, \dots\}$

$$3) (aba)^+ + (a^+b^+)^+$$

$$(a^+b^+)^+ = (a+b)^+$$

$$(aba)^+ = \{ \epsilon, aba, abaaba, \dots \}$$

$$4) (i) [a-zA-Z][a-zA-Z0-9]^+$$

↓
Identifier
must start
with letter.

↓
This allow for zero or
more subsequent letter or
digits.

(ii) Priority Rule :

→ In Lex, if 2 pattern match the same input string, the pattern list first takes precedence.

Longest match rule :-

→ The lexical analyzer will consume longest possible string that matches a pattern.

numeric formats :-

→ Distinguish b/w valid & invalid number formats.

$$(5) (i) (0^+1^+)^+ = (0+1)^+$$

$$(0^+1^+)^+ = \{ \epsilon, 0, 1, 00, 01, 10, 11, \dots \}$$

$$(0+1)^+ = \{ \epsilon, 0, 1, 00, 01, 10, 11, \dots \}$$

$$(0^+1^+)^+ = (0+1)^+$$

$$(ii) (0+1)^+ (0+1)^+ = (0+1)^+$$

$$R^+ R^+ = R^+$$

$$(ii) \quad (O^*)^* = O^*$$

$$(R^*)^* = R^*$$

$$(iv) \quad (O+1)^* O (O+1)^* + (O+1)^* 1 (O+1)^* = (O+1)^*$$

$$(O+1)^* O + (O+1)^* 1 = (O+1)^*$$

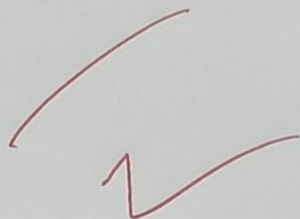
$$(O+1)^* (1+O) = (O+1)^*$$

$$(O+1)^* = (O+1)^*$$

$$(v) \quad E + O(O)^* = O^*$$

$$E + R(R^*) = R^*$$

~~QED~~ ~~QED~~



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation